

Leveraging LLMs for Data Analysis in Education Research

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Stanford

SCALE Initiative
Accelerator for Learning

About me

- Stanford Math B.S. (2023), University of Michigan Statistics M.A. (2025)
- Senior Research Data Analyst here at SCALE (2025-now)
- I use natural language processing (NLP) to study conversational data
- I study what moves tutors use in sessions to engage and teach students
- Would love to be a resource for Stanford or post-Stanford life!

What we'll do today

The plan

I want to show you **two concrete ways** to use LLMs when working with data: **pre-processing** and **analysis**.

My goal

You'll come away with both *ideas* for how to use LLMs, some *intuition* for why using LLMs requires special attention in research, and *experience* using LLMs to process and analyze educational data.

Roadmap

- **1. Preprocessing**
- 2. Pitfalls
- 3. Analysis
- 4. Pitfalls
- 5. Takeaways

Preprocessing: from files to tables

- Often data comes to us as a series of **unstructured files**.
- The data analysis procedures we work with typically want something closer to a **table-like structure**.
- So we're going to dive into turning **this** (1) into **this** (2).

Name	Date Modified
TalkBack.Year1.Baker.Fall.121019-1.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Baker.Fall.121019-2.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Baker.Spring.012120.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Baker.Spring.013120.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Baker.Spring.022820.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Barry.Spring.011420-1.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Barry.Spring.011420-2.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Barry.Spring.011520.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Barry.Spring.013120.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Barry.Spring.021120.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Barry.Spring.030220-1.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Barry.Spring.030220-2.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Benson.Fall.121019.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Benson.Spring.011320.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Benson.Spring.011420.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Benson.Spring.011620.xlsx	Feb 5, 2026 at 2:57 PM
TalkBack.Year1.Benson.Spring.020520.xlsx	Feb 5, 2026 at 2:57 PM



State	Reading	Math	Writing	Sat	Ratio	Drop
Iowa	610	615	588	3	6.8	2.3
Minnesota	595	609	578	7	7.7	3.0
Missouri	595	600	584	5	6.9	3.7
Wisconsin	594	608	582	5	8.3	2.2
N. Dakota	590	593	566	3	6.2	2.3
S. Dakota	589	600	569	3	7.1	3.9
Illinois	588	583	583	6	4.0	4.0
Nebraska	587	594	572	4	6.7	2.8
Michigan	584	603	575	5	8.1	7.4
Kansas	581	589	564	7	8.6	2.7

Preprocessing: the data

- **MIT OpenCourseWare** is a website where you can view MIT classes and course materials online.
- These are undergraduate and graduate courses from various departments at MIT.
- I've scraped a **random sample of 100 syllabi** from MIT OCW.



Preprocessing: your task

Goal

Build and run a prompt that extracts some **structured information** from these syllabi, then report what you find.

Some examples of structured information to extract:

- Does the course have any **prerequisites**? (True / False)
- What percent of the grade comes from **exams** / **homework** / **reading** / **participation**? (number)
- How many **required readings** does the course have? (number)

Focus

Extract a well-defined **True/False** value or a **number** from each syllabus.

Preprocessing: your turn

Go to `https://julian-bernado.github.io/llms4edu/`

Example research questions

- Do courses with prerequisites have more required readings?
- How is grade weight distributed across departments?
- What share of courses have a final exam?

Tips

- Ask for one well-defined field at a time.
- Tell the model exactly what output format you want.
- Spot-check a few syllabi by hand.

I'll be walking around — flag me down with questions.

Preprocessing: discussion

Open-ended

What did you find? Any reactions to your results?

Directed questions

- Were any of your results **surprising**?
- Did any people measuring the **same thing** get **different results**?

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Pitfall #1: Reproducibility

Different people can run “the same” analysis and get different numbers. Three common reasons:

1. **Different models**

Results depend on which model you used — be transparent about model choice.

2. **Benign prompt differences**

Small wording changes can shift results — track exactly which prompt your results came from.

3. **Sampling**

Models are stochastic by default — set `temperature = 0` for more deterministic output.

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Analysis: interpreting data

- **Preprocessing** uses LLMs to *extract* structured information.
- **Analysis** uses LLMs to *interpret* data.
- In reality, the boundary between these two tasks is not always so clear.
- Now we're going to use LLMs to help us **understand** our data.

Analysis: the data

- These are transcripts from **classroom conversations** (the **TalkMoves** dataset).
- A common test dataset in the education NLP space.
- **K–12 Mathematics** lessons.

Table 2: Sample annotated excerpt from a classroom session

Speaker	Sentence	Teacher Tag	Student Tag
Teacher	Look, we have a different model over here, even.	1 - None	
Teacher	So now we have three.	1 - None	
Teacher	I wonder if it's going to be the same as yours, or if it's going to be the same as this one.	1 - None	
Teacher	Is two thirds still bigger, Greg, is two thirds still bigger than a half, on this model too, or did it change?	8 - Press for Accuracy	
Teacher	Ok, Danielle, what do you think about this time?	8 - Press for Accuracy	
Student	Well, um, two thirds		4 - Making a Claim
Teacher	What is two thirds?	8 - Press for Accuracy	
Teacher	Can you build a two thirds and a one half for him separate so we can then compare?	8 - Press for Accuracy	

Analysis: your task

Goal

Imagine you want to understand some **behavior** in these classroom conversations. Write up a description of a behavior — in teachers or students — that might be present in the transcripts.

Some examples of questions to answer:

- **Praise** after getting a problem correct
- Student **confusion** / **frustration**
- **Classroom management**

Analysis: your turn

Go to `https://julian-bernado.github.io/llms4edu/`

Prompting tips

- **Give examples** of the behavior you're looking for.
- Give as much **context** as possible for what you're trying to do.
- **Read** some transcripts to understand what the conversations look like.

I'll be walking around — flag me down with questions.

Analysis: discussion

Open-ended

What types of things did people look for?

Directed questions

- Did anybody **disagree** with the examples the LLM pulled out?
- If so, how would you **modify your prompt** knowing how it extracted things?
- If you were reading research studying these transcripts, and the outcome was **annotated by an LLM** this way — what problems do you foresee? What disclaimer would be appropriate? What sources of **bias** can we imagine?

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Pitfall #2: Validity

- LLMs can quickly generate **all** of these annotations.
- But how do we know the LLM was **right**?

Things to think about doing

- **Human validation rounds**
- Measuring **high agreement** (human vs. model)
- Writing **unambiguous codebooks**

Pitfall #3: Privacy & pricing

- When we use these LLMs, we're calling an **API** and sending our data to a company's servers (OpenAI / Anthropic / Google).
- For **private data** — which some of you may be working with — this can't be freely sent to a company's servers.

Routes you might use LLMs from

- **Direct from an API** — convenient, but data leaves your control.
- **Sherlock** (or other compute clusters) — institutional, more control.
- **Locally** — data never leaves your machine.

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Takeaways

Two use cases

1. Preprocessing
2. Analysis

Three pitfalls

1. Reproducibility
2. Validity
3. Privacy & pricing

Closing question

Can you imagine using LLMs in your work this summer, or in the stuff you're interested in? What kinds of questions could it let you answer? (All good if not!)

Thanks!

A decorative graphic on the right side of the slide, consisting of several concentric circles. Each circle is composed of many small, rectangular segments in various shades of blue and teal, creating a mosaic-like effect.